Alexandra Schuller, CPI

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SKILLS AND QUALIFICATIONS

Languages Bilingual in French and English, with native-level proficiency in both.
Modelling and Simulation | CSI Bridge • SAP2000 • Abaqus • DIANA • ETABS • SAFE • Revit • AutoCAD • HEC-RAS • SUMO • WoodWorks
Standards | CSA S6 • CSA S7:23 • CSA A23.3 • CSA S16 • CSA O86 • NBCC 2020
Programming | Python • MATLAB • VBA (Excel) • Octave • GitHub • Claude Code • WordPress
Project Management | MS Project • Primavera P6 • Procore • Aconex • Bluebeam • Stack • Asana • Notion
Other | Accpac/Sage • Concur • Windows • Linux • MacOS
Licences & Certifications | Driver's licence • ASP Construction Health and Safety (2022) • ASP Lockout • WHMIS for laboratory personnel (Concordia)

EDUCATION

Bachelor of Engineering, Civil Engineering (Infrastructure Specialisation), with Distinction **2022-2026**
Co-operative Education Program (Co-op)
Concordia University, Montreal, QC
Member of the Institute for Co-operative Education

University Prerequisite Program for Adults, Continuing Education **2021-2022**
Dawson College, Semester Honours

Bachelor of Arts, Classical Studies, Specialization in Archaeology **2009-2013**
McGill University, Montreal, QC

DEC in Arts and Sciences **2007-2009**
Marianopolis College, Montreal, QC

DESIGN PROJECTS

CapDécarie, Civil Engineering Capstone Project **September 2025 - April 2026**
Took the initiative to propose a project outside the options offered, then recruited and rallied a team of five around the concept: covering a section of the Décarie Expressway trench, between Snowdon and Queen Mary, with a 5,670 m^2 pedestrian park, and replacing the aging Queen Mary overpass.

- Responsible for the design of the foundations and beams of the cap structure (a deck of about 180 m), the drainage design and the traffic simulation for construction staging.
- Redesigned the Queen Mary overpass and validated structural behaviour with finite element analysis (CSI Bridge, SAP2000); steel and reinforced concrete design, load analysis and seismic analysis.
- Also covered embodied carbon, cost estimating and construction planning.

Rock Slope Stability Analysis, Northern Quebec **January - April 2026**
Course project, Advanced Geotechnical Engineering (CIVI 437), Concordia University
Stability analysis from triaxial test data using the Hoek-Brown failure criterion, converted to equivalent Mohr-Coulomb parameters (Excel, MATLAB), and finite element modelling in Abaqus; assessed the effect of GSI (100 and 85) on cohesion, friction angle and stability.

10-Storey Reinforced Concrete Office Building, Quebec City **September - December 2025**
Course project, Design of Reinforced Concrete Structures (CIVI 453), Concordia University
Complete structural design of a reinforced concrete office building (flat slabs, columns and ductile shear walls) to NBCC 2020 and CSA A23.3: wind and seismic loads, modelling in ETABS and SAFE.

Captain, Concordia Concrete Canoe Team **September 2023 - May 2026**
Revived the team after seven years of inactivity and led twelve people in designing, building and transporting a 6-metre concrete canoe to the national competition in Quebec City.

- Concrete mix design, laboratory testing and structural analysis of the canoe.

WORK EXPERIENCE

Intern, Direction Générale des barrages **April - August 2025**
Major Projects Directorate
Ministry of the Environment, the Fight Against Climate Change, Wildlife and Parks (Quebec)
Helped implement a new centralized MS Project planning system for about 1,000 dams in Quebec by consolidating project schedules and milestones with Python and AI tools.

Intern, Renouveau Lafontaine - Louis-Hippolyte-La Fontaine Tunnel **January - April 2025**
Pomerleau
Tracked the progress of heating cable repairs in the main and access tunnels; built a Python tree of the tunnel's electrical systems that automatically generates their power supply diagrams.

Intern, Local Energy Efficiency Partnerships (LEEP) **September - December 2024**
Natural Resources Canada
CanmetENERGY, Ottawa
Designed a VBA data analysis tool (costs, GHG, energy) for converting homes to heat pumps, which uncovered an error in published government data and led to a revision.

Program Assistant - The McConnell Foundation **January 2015 - August 2021**
Montreal, QC
Supported the Cities for People program, which funds projects that improve urban life across Canada.

Intern & Research Assistant - Université de Montréal **Summers 2011 and 2012**
Archaeological excavations at *Argilos*
Surveyed the site, laid out trenches and documented the excavations following archaeological protocols.

AWARDS & DISTINCTIONS

Graduated with Distinction - Bachelor of Engineering, Civil, Concordia University **June 2026**

Pierre Sauvé, Eng. Scholarship for the Advancement of Women in Engineering **March 2023**
Fondation de l'Ordre des ingénieurs du Québec (FOIQ)

Valedictorian - Centennial Regional High School (CRHS) **June 2007**

PROFESSIONAL ASSOCIATIONS

Candidate to the Engineering Profession (CPI) - Ordre des ingénieurs du Québec **September 2026**

Board Member **December 2022 - May 2023**
The Concordia Greenhouse, Montreal, Quebec
Member of the infrastructure committee overseeing options to upgrade the greenhouse's existing components.

OTHER INTERESTS

Sports	Hockey, softball and tennis; Triathlon Montréal Esprit Sprint (2020)
Other Passions	Greenhouse design, archaeology, community gardens, music (200+ public performances)